

**Ex No:9**

## **SQLite BANK DATABASE**

**DATE:**

**AIM:**

To implement the SQLite bank database in android application.

**ALGORITHM:**

STEP 1: Create a class that extends SQLiteOpenHelper to manage the creation and version management of the database.

STEP 2: Implement methods in your application to perform database operations using SQLiteDatabase class.

STEP 3: You can then use this BankDataSource class in your Android activities

STEP 4: Use SQL commands to create the database schema.

STEP 5: Create tables for each entity with the appropriate attributes.

STEP 6: Apply normalization techniques to organize the data efficiently.

## CODE: XML

```
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/editTextAccountNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Account Number"
        android:layout_marginTop="16dp"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="16dp"/>
    <EditText
        android:id="@+id/editTextBalance"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Balance"
        android:layout_below="@id/editTextAccountNumber"
        android:layout_marginTop="16dp"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="16dp"/>
    <EditText
        android:id="@+id/editTextAccountHolder"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Account Holder"
        android:layout_below="@id/editTextBalance"
        android:layout_marginTop="16dp"
```

```

        android:layout_marginStart="16dp"
        android:layout_marginEnd="16dp"/>
<Button
    android:id="@+id/buttonAddAccount"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add Account"
    android:layout_below="@id/editTextAccountHolder"
    android:layout_marginTop="16dp"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"/>
</RelativeLayout>

```

CODE: MainActivity.java

```

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.content.ContentValues;
import android.content.Context;
import android.database.SQLException;
import android.database.sqlite.SQLiteDatabase;
public class BankDatabaseHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "bank.db";
    private static final int DATABASE_VERSION = 1;
    // Table name and column names
    public static final String TABLE_ACCOUNTS = "accounts";
    public static final String COLUMN_ID = "_id";
    public static final String COLUMN_ACCOUNT_NUMBER = "account_number";
    public static final String COLUMN_BALANCE = "balance";

```

```

    public static final String COLUMN_ACCOUNT_HOLDER = "account_holder";
// SQL statement to create the accounts table

    private static final String DATABASE_CREATE = "create table " +
TABLE_ACCOUNTS + " ("
        + COLUMN_ID + " integer primary key autoincrement, "
        + COLUMN_ACCOUNT_NUMBER + " text not null, "
        + COLUMN_BALANCE + " real not null, "
        + COLUMN_ACCOUNT_HOLDER + " text not null);";

public BankDatabaseHelper(Context context) {
    super(context, DATABASE_NAME, null, DATABASE_VERSION);
}

@Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(DATABASE_CREATE);
    }

@Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_ACCOUNTS);
        onCreate(db);
    }
}

public class BankDataSource {
    private SQLiteDatabase database;
    private BankDatabaseHelper dbHelper;

    public BankDataSource(Context context) {
        dbHelper = new BankDatabaseHelper(context);
    }

    public void open() throws SQLException {
        database = dbHelper.getWritableDatabase();
    }
}

```

```
public void close() {
    dbHelper.close();
}

public long createAccount(String accountNumber, double balance, String
accountHolder) {
    ContentValues values = new ContentValues();
    values.put(BankDatabaseHelper.COLUMN_ACCOUNT_NUMBER,
accountNumber);
    values.put(BankDatabaseHelper.COLUMN_BALANCE, balance);
    values.put(BankDatabaseHelper.COLUMN_ACCOUNT_HOLDER,
accountHolder);
    return database.insert(BankDatabaseHelper.TABLE_ACCOUNTS, null,
values);
}

BankDataSource dataSource = new BankDataSource(this);
dataSource.open();
long accountId = dataSource.createAccount("1234567890", 1000.00, "John Doe");
dataSource.close();
}
```

## OUTPUT

Account Number

Balance

Account Holder

Add Account

The image shows a mobile application interface for adding a new account. It features a light pink background with a dark blue header and a dark blue bottom bar. The form consists of three text input fields labeled 'Account Number', 'Balance', and 'Account Holder'. Below these fields is a purple button with the text 'Add Account'. The entire form is enclosed in a dark blue border.

**RESULT:**

Thus, The Android Application has been developed and run successfully.